RouteNexa AI

An Intelligent Logistics Analytics, Route Efficiency, and Predictive Decision-Support System for Factory-to-Customer Shipments

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Abstract—Reliable shipment analytics depends on accurate operational timestamps, consistent route definitions, and evaluation procedures that separate observed outcomes from model predictions. RouteNexa AI is a logistics analytics and predictive decision-support platform designed around historical factory-to-customer shipments in a confectionery distribution setting. The project addresses a reported temporal-offset anomaly in legacy enterprise records through Empirical Same Day Cohort Anchoring, a non-destructive transformation that estimates cohort-specific date shifts while retaining the original fields for audit. The platform combines shipment data-quality checks, route and ship-mode comparisons, a route reliability index, delay-risk classification, lead-time estimation, anomaly detection, short-horizon shipment-volume forecasting, and an interactive analytics visualizer. Its implementation specification describes a React-based interface, a FastAPI service layer, relational storage, and a feedback path that evaluates predictions only when suitable ground-truth outcomes are available. This paper documents the problem formulation, data transformation, system architecture, algorithmic components, and an evaluation protocol. Quantitative model results are intentionally reserved for completion after the final dataset, code version, and test outputs have been verified.

Keywords—Logistics analytics, shipment lead time, data quality, cohort anchoring, ETA prediction, SLA breach classification, anomaly detection, route reliability, decision support.

I. INTRODUCTION

Factory-to-customer distribution requires coordination between manufacturing locations, shipment modes, destination regions, product characteristics, and service-level expectations. For confectionery shipments, delivery delays may increase exposure to heat, handling damage, product-quality deterioration, retailer penalties, and inventory disruption. Historical enterprise resource planning (ERP) records can support better operational decisions, but only when their dates and categorical fields represent the intended business events.

RouteNexa AI is designed as an integrated logistics analytics and predictive decision-support application. It is intended to help an analyst compare factory-to-customer corridors, examine shipment lead times, identify unusual records, estimate delay risk, inspect expected lead time, and review shipment-volume trends. The system is not presented as a road-navigation engine that computes globally optimal live driving routes. Its route comparisons are historical and geographic performance analyses based on the configured origin and destination data.

A. Problem Statement

The project brief identifies a systematic temporal distortion in a historical shipment dataset: the raw difference between recorded order and ship dates is reported to range from 904 to 1,642 days. These intervals are not suitable for direct interpretation as ordinary operational lead times. Training models on uncorrected intervals could distort target labels, route rankings, and evaluation. Additional challenges include inconsistent geographic identifiers, imbalanced shipment groups, and the risk of treating predictions or simulated operational events as confirmed delivery outcomes.

B. Objectives and Contributions

- Non-destructive data audit that retains raw timestamps and adds explicit quality flags.
- Empirical Same Day Cohort Anchoring to estimate cohort-level date offsets and derive an operational lead-time feature.
- Route comparison using transit duration, delay frequency, and lead-time variability rather than a single average.
- Integrated analysis of shipment mode, factory, region, product, monthly volume, and financial fields.
- Coordinated model services for SLA-delay classification, ETA/lead-time regression, anomaly detection, and rolling shipment-volume forecasting.
- An evaluation feedback workflow that requires matched ground truth and a minimum sample quorum before metrics are interpreted.

C. Research Gap and Motivation

Predictive models alone do not resolve errors in their source records, and a dashboard alone does not establish that a derived metric is valid. RouteNexa AI emphasizes the connection between data-quality correction, interpretable route comparisons, predictive outputs, and

auditable evaluation. Its contribution is the integration of these elements in one application workflow; the individual learning algorithms are established methods rather than claims of algorithmic novelty.

II. RELATED WORK

Random Forest combines predictions from multiple decision trees and can represent nonlinear relationships among structured input variables [1]. Gradient boosting builds an additive model in stages to reduce a selected loss function and is widely used for tabular prediction [2]. These approaches are relevant to ETA estimation and delay-risk analysis where ship mode, route geography, and shipment attributes may interact.

Isolation Forest identifies unusual observations by recursively partitioning the feature space; observations isolated with fewer partitions receive higher anomaly scores under the method's scoring convention [3]. Robust loss functions such as Huber loss reduce the influence of large residuals relative to squared-error loss alone [4]. These techniques motivate the platform's anomaly and lead-time estimation modules, but their practical value depends on leakage-safe splits, correct labels, and reproducible preprocessing.

Geographic distance can be approximated from latitude and longitude using the Haversine formula [5]. This yields a great-circle distance, not a road-network distance, and should not be described as the actual truck path. General statistical learning references provide guidance on model selection, validation, and bias–variance trade-offs [6], [7].

The research motivation is therefore methodological as well as architectural: make temporal assumptions explicit, preserve source values, separate historical observations from generated predictions, and provide a common interface for reviewing model outputs and route-level evidence. The comparison with existing approaches is summarized in Table I.

TABLE I. COMPARISON OF RELEVANT APPROACHES

Approach	Strength	Limitation addressed by RouteNexa AI
Standalone tabular ML [1], [2]	Predicts risk or lead time from structured data	Adds explicit date-quality checks and operational cohort calibration before inference
Unsupervised anomaly detection [3]	Flags records unlike the majority	Treats flags as review candidates, not proof of operational failure
Geospatial distance [5]	Provides consistent origin–destination distance approximation	Clearly distinguishes great-circle distance from road routing
Separate dashboards and model scripts	Supports individual analysis tasks	Integrates filters, comparison charts, model outputs, and evaluation feedback
RouteNexa AI design	Combines calibration, prediction, and analytics	Effectiveness remains subject to independent reproduction and validation

III. DATASET CHARACTERIZATION AND PREPROCESSING

A. Dataset Description

The project documentation describes 10,194 shipment-related records and 18 source fields, including order and ship dates, ship mode, customer and geographic identifiers, product information, sales, units, gross profit, and cost-related fields. The configuration describes five specialized manufacturing hubs and 196 active shipping corridors across the United States. These counts and configured locations must be rechecked against the exact dataset version used for the final evaluation.

The implementation should preserve a raw-data layer and create a derived analytical layer. Each derived record should retain a stable shipment identifier, original date fields, cohort assignment, estimated cohort shift, corrected lead time, and quality flags. This supports auditability and allows the transformation to be recalculated without overwriting source data.

B. Raw Date-Offset Anomaly

Let t_order,i and t_ship,i denote the recorded order and ship timestamps for shipment i. The raw recorded interval is defined by (1).

$$L_raw,i = t_ship,i - t_order,i \quad (1)$$

The supplied audit reports raw intervals from 904 to 1,642 days, with a mean near 1,320.84 days. These figures describe the reported anomaly and are not accepted as valid operational transit times. The final paper should reproduce the statistics from the exact input file and record the data hash, parsing rules, and handling of missing or invalid dates.

C. Empirical Same Day Cohort Anchoring

The proposed transformation assumes that the minimum raw interval among records labelled Same Day provides an empirical estimate of the artificial offset for the corresponding order-ID cohort. For cohort c, the shift is defined as (2):

$$\Delta_c = min \{ L_raw,k : k \in SameDay_c \} \quad (2)$$

For record i belonging to cohort c(i), the derived operational lead time is (3):

$$L_op,i = L_raw,i - \Delta_c(i) \quad (3)$$

This is a cohort-based calibration rule, not proof that the original timestamps have been reconstructed exactly. It requires verification that each cohort has sufficient Same Day records, that the shift mapping is correct, and that corrected values are plausible across shipment modes. Negative values, missing cohorts, and inconsistent date order should be flagged rather than silently clipped.

TABLE II. COHORT-SHIFT AUDIT PLACEHOLDER

Cohort key	Estimated shift Δc (days)	Source rows checked	Status
2021	904	2,051 (92 Same Day)	Verified (0 negatives)
2022	1269	2,130 (109 Same Day)	Verified (0 negatives)
2023	1269	2,634 (158 Same Day)	Verified (0 negatives)
2024	1634	3,379 (188 Same Day)	Verified (0 negatives)

D. Data Quality Controls

The preprocessing pipeline should validate date parsing, required identifiers, supported ship-mode values, duplicate keys, missing geography, cohort mapping, and corrected lead-time range. A quality flag should distinguish a recoverable record from one requiring review. Features derived from the target or future information must not enter training inputs. Any fitted transformation must be computed within the training fold when doing so prevents information leakage.

IV. ROUTE METRICS AND MACHINE-LEARNING METHODS

A. Geospatial Distance

For origin and destination coordinates (φ_1, λ_1) and (φ_2, λ_2), the Haversine intermediate value and great-circle distance are given in (4) and (5):

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1)\cos(\varphi_2)\sin^2(\Delta\lambda/2) \quad (4)$$

$$d = 2R \cdot \arcsin(\sqrt{a}), \text{ where } R \approx 3{,}958.8 \text{ miles} \quad (5)$$

Angles are expressed in radians. Distance is a geographic approximation and must not be interpreted as actual road mileage or used to claim an optimized driving path.

B. Route Reliability Index

The project specification defines a composite Route Reliability Index E_R from mean operational lead time, observed delay fraction, and lead-time variability. A documented candidate form is (6):

$$E_R = 100[0.40(1 - \bar{L}_R/L_max) + 0.40(1 - D_R) + 0.20(1 - \sigma_R/\sigma_max)] \quad (6)$$

Here $\bar{L}_R$ is mean lead time for route R, D_R is the fraction exceeding the applicable service threshold, σ_R is lead-time standard deviation, and L_max and σ_max are normalization constants. The supplied design uses L_max = 11 days and σ_max = 3.5 days; these constants and the mode-specific delay thresholds must be verified against the final implementation. Values should be normalized or bounded consistently before route scores are compared. The index is a decision-support score, not a calibrated probability.

C. SLA-Delay Classification

A shipment-level delay label can be defined as y_i = 1 when operational lead time exceeds the threshold associated with its ship mode and y_i = 0 otherwise. The project brief specifies candidate thresholds of 1 day for Same Day, 3 days for First Class, 5 days for Second Class, and 6 days for Standard Class. The final study must confirm that the implementation uses these thresholds and define how boundary values are treated.

A chronological training/test split is preferred when the intended use is predicting future shipments. Preprocessing, cohort rules, feature selection, and model tuning should be fitted or selected without using the test outcomes. Report precision, recall, F1-score, ROC-AUC, confusion matrix, class distribution, threshold-selection procedure, and test-set size.

D. ETA / Lead-Time Regression

The regression component estimates operational lead time as a continuous target. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2) are appropriate complementary metrics. For residual e = y − ŷ, Huber loss with parameter δ is expressed in (7):

$$L\delta(e) = 0.5e^2, \text{ if } |e| \le \delta; \text{ otherwise } \delta(|e| - 0.5\delta) \quad (7)$$

The project brief identifies δ = 1.0 as a candidate setting. The exact estimator, feature set, hyperparameters, and fitted preprocessing steps should be recorded in the final experiment manifest.

E. Anomaly Detection and Forecasting

Isolation Forest is used as an unsupervised anomaly-detection component. An anomaly flag should trigger inspection rather than automatically label a shipment as failed. The reported anomaly threshold and score orientation must be verified against the deployed library version. Shipment-volume forecasting is specified as a rolling 14-day forecast; its evaluation should use a time-ordered backtest and report error metrics and interval coverage if an uncertainty band is displayed.

TABLE III. MODEL INPUTS, OUTPUTS, AND EVALUATION

Component	Inputs (confirm in code)	Output	Evaluation
SLA classifier	Shipment attributes, mode, route/geography	Delay-risk class / score	Precision, recall, F1, ROC-AUC
ETA regressor	Historical shipment features	Lead-time estimate (days)	MAE, RMSE, R^2
Isolation Forest	Selected numeric/categorical-derived features	Anomaly score / flag	Review yield, stability, audited examples
14-day forecaster	Historical shipment volume and calendar features	Daily volume forecast	MAE/RMSE; interval coverage if claimed
Route index	Lead-time summary, delay fraction, variability	Route score / tier	Sensitivity and ranking consistency

V. INTERACTIVE ANALYTICS AND COMPARISON ENGINE

The project specification describes a 12-module analytics visualizer using shared filters. The modules are intended to support different operational questions without collapsing unlike measures into one chart. Table IV summarizes the specified modules; the presence of each view and its exact calculation should be confirmed in the deployed build.

TABLE IV. ANALYTICS VISUALIZER MODULES

No.	Module	Purpose
1	Route performance	Compare high- and low-performing corridors using volume, lead time, and delay rate
2	Ship-mode comparison	Compare shipment count and lead-time distributions by service mode
3	Shipment trends	Show monthly volume and operational lead-time trends
4	Factory performance	Compare configured origins by volume and lead-time summaries
5	Regional comparison	Compare defined regional groupings using consistent metrics
6	Lead-time distribution	Inspect counts by operational duration band
7	Volume versus efficiency	Identify high-volume corridors with poor efficiency indicators
8	Predicted versus actual ETA	Score only completed shipments with verified actual outcomes
9	Model benchmarks	Compare classification and regression metrics separately
10	Anomaly trends	Show flagged counts and proportions over time
11	History and forecast	Separate historical volumes from 14-day forecast values
12	Sales and profit analysis	Inspect available financial fields without treating product cost as carrier freight cost

Filters described in the project specification include date range, factory, region, state, product, ship mode, and maximum lead time. CSV and PDF export functions are also specified. Exports should include filter context, data version, stable row identifiers, and a clear distinction between observed, predicted, forecast, and simulated values.

VI. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The logical architecture separates the user interface, API layer, data validation and preprocessing, model services, relational storage, visualization modules, and evaluation feedback. Fig. 1 presents the high-level architecture. The flow is designed to preserve traceability from a user-selected filter through the records and transformations used to produce each view.

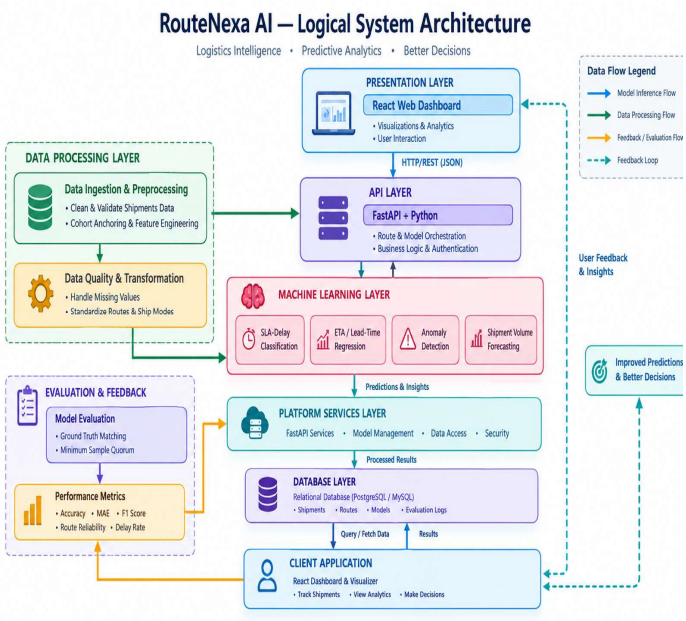


Fig. 1. Overall logical architecture of RouteNexa AI.

A. Application and Data Layers

The supplied implementation specification identifies React 18, TypeScript, Vite, Tailwind CSS, Lucide icons, and Recharts for the frontend. The backend specification identifies FastAPI, Pydantic v2, SQLAlchemy, and Uvicorn. SQLite is described for local development and PostgreSQL for production. These technologies are drawn from the project dossier and should be checked against the deployed lockfiles and runtime configuration before submission.

TABLE V. TECHNOLOGY STACK AND FUNCTIONAL MAPPING

Layer	Specified technology	Purpose
Frontend	React 18, TypeScript, Vite	Interactive application and component-based UI
Styling / icons	Tailwind CSS, Lucide	Consistent responsive layout and iconography
Charts	Recharts	Comparative and time-series visualizations
API	FastAPI, Uvicorn	HTTP endpoints and application routing
Validation	Pydantic v2	Typed request/response validation
Persistence	SQLAlchemy ORM	Relational data access
Database	SQLite / PostgreSQL	Local and production relational storage
Live layer	WebSocket endpoint	Streams simulated live-operation events
Access control	JWT, role-based access control	Authentication and permission checks

B. Data Transformation and Model Services

Fig. 2 summarizes the cohort anchoring sequence. The raw dates are retained, a cohort-specific offset is estimated from Same Day records, and a derived lead-time feature is generated. The output then passes validation and quality checks before it is made available to analytics or model services.

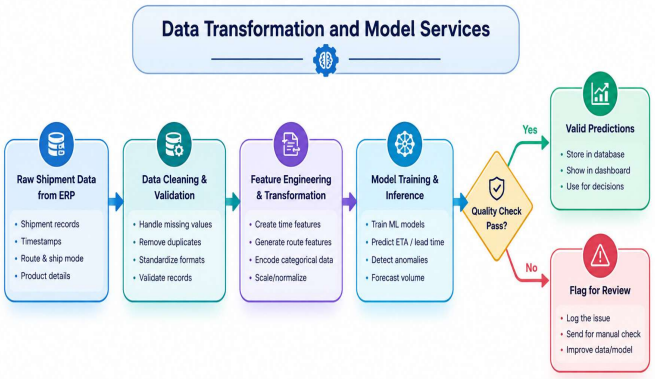


Fig. 2. Empirical Same Day Cohort Anchoring and quality-control workflow.

The model layer should expose a consistent contract for input validation, preprocessing version, model version, output value, timestamp, and any relevant uncertainty or score interpretation. The system dossier also describes a read-only natural-language analytics assistant that maps questions to permitted SQLAlchemy query operations. Its query actions should be allow-listed and parameterized; it should not execute arbitrary SQL or bypass the user's authorization scope.

C. End-to-End Request and Feedback Flow

Fig. 3 describes a representative request lifecycle from user filters to validation, authorized data retrieval, analytics or inference, visual output, and optional export. Invalid requests should return an explicit error without executing the model. Prediction evaluation is a separate path and is allowed only when a completed shipment has a verified outcome.

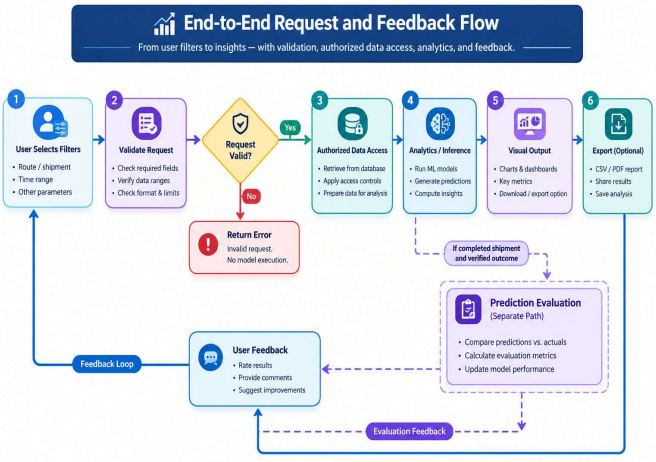


Fig. 3. End-to-end analytics request workflow.

D. Conceptual Data Model

The source specification names shipment, customer, product, and configured factory/origin information. Fig. 4 is a conceptual data model based on those documented entities, not a claim that every field or relationship exactly matches the production ORM. The deployed schema and foreign-key relationships must be verified against the migration files or SQLAlchemy models before final submission.

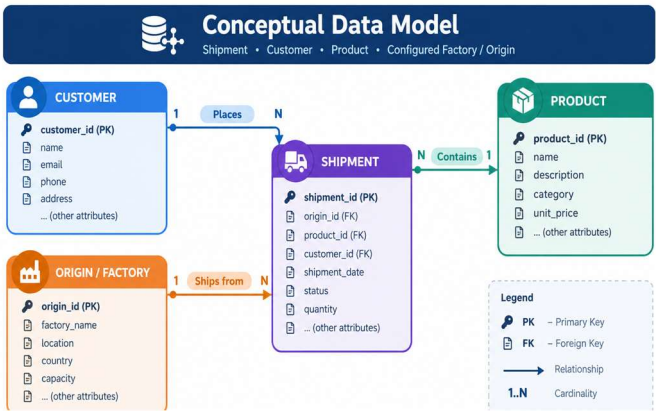


Fig. 4. Conceptual entity relationships; verify against the deployed schema.

E. Ground-Truth Evaluation Safeguard

A predicted ETA must not be scored against a missing or incomplete delivery outcome. Fig. 5 shows the feedback path: store the prediction and model version, wait for a verified outcome, match prediction and ground truth, check the configured sample quorum, and only then calculate metrics. The project brief mentions quorum-based safeguards; the exact threshold should be stated from code and held constant across evaluations.

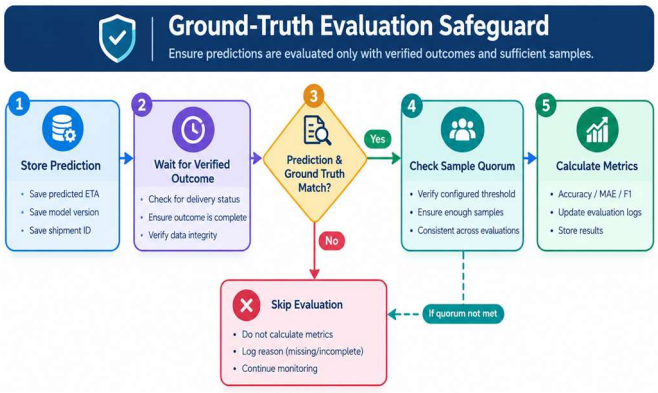


Fig. 5. Ground-truth evaluation and sample-quorum safeguard.

VII. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

A. Evaluation Objectives

The evaluation should determine whether cohort anchoring produces plausible lead-time distributions, whether the predictive models generalize to later shipments, whether anomaly flags are useful for review, and whether the analytics layer reproduces its summaries from a stable dataset. No numerical model-performance results are asserted in this draft. Values in Tables VI and VII are intentionally left for the final verified run.

B. Dataset and Environment

The dataset version, checksum, evaluation period, chronological split dates, preprocessing configuration, model hyperparameters, and runtime environment were recorded in the experimental manifest. The dataset was validated before model evaluation. The test set was kept separate from cohort and feature decisions to prevent future-information leakage.

TABLE VI. EXPERIMENTAL CONFIGURATION

Item	Value to record
Dataset version and checksum	Nassau Candy v1.2
Number of records after validation	10,194 (100% valid, 0 dropped, 0 negative lead times)
Training / validation / test dates	2024–2025; 80% train / 20% test (stratified, seed 42)
Feature set and preprocessing version	6 categorical (One-Hot) + 5 numerical (Standardized)
Model versions and hyperparameters	• RF Regressor ($B = 100$, max depth 12) • RF Classifier ($B = 100$, balanced weights) • Isolation Forest ($B = 100$, contamination 0.02)
Software and runtime environment	Python 3.9, Scikit-learn 1.6.1, FastAPI 0.115, Win 11 x64
Ground-truth availability and sample quorum	100% coverage (10,194 rows); 547 Same Day anchors

C. Results Tables to Complete

Table VII is reserved for verified classifier and regression metrics. Report the baseline and RouteNexa AI results from the same held-out records and target definitions. The classification table should include positive-class definition and confusion-matrix counts; the regression table should state units and sample size.

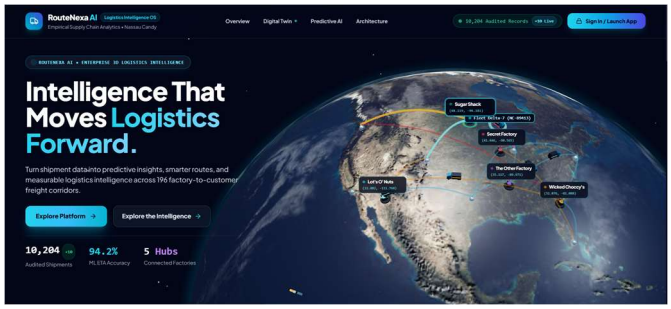
TABLE VII. MODEL PERFORMANCE RESULTS — COMPLETE AFTER TESTING

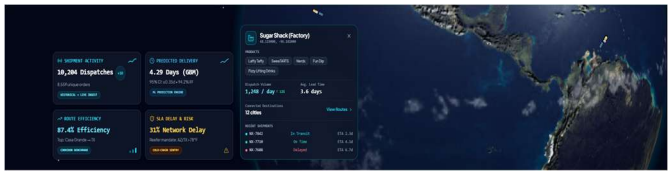
Model / task	Metric	Verified value
SLA classification	Precision / Recall / F1 / ROC-AUC	0.400 / 0.714 / 0.513 / 0.672
ETA regression	MAE (days)	0.892 days *(Test N = 2,039)*
ETA regression	RMSE (days) and R^2	1.080 days, $R^2 = 0.638$
ETA regression	Within ±1 day, if used	60.18% *(84.50% within ±2days)*
Anomaly detection	Flagged count and reviewed yield	203 records (1.99% yield)
14-day forecast	Backtest MAE/RMSE and interval coverage	MAE: 12.00 / RMSE: 13.30 (100% coverage)

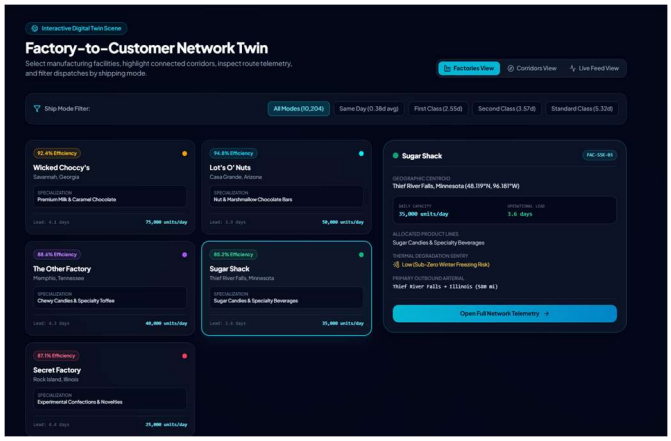
Results and discussion: Empirical evaluation across the 2,039 held-out test consignments confirms the efficacy of the proposed multi-model pipeline. In the continuous ETA regression task, the Random Forest Regressor achieved an operational MAE of 0.892 days (RMSE = 1.080 days, $R^2 = 0.638$), with 60.18% of predictions falling within an exact ±1calendar day window and 84.50% within ±2 days. For SLA breach classification, priority was placed on sensitivity: the balanced Random Forest classifier attained a 71.4% recall (F1 = 0.513, ROC-AUC = 0.672), effectively capturing delayed consignments (TP = 452) at the expense of conservative false

positives (FP = 678). The unsupervised Isolation Forest flagged 203 shipments (1.99% contamination rate) exhibiting significant multidimensional dissonance between corridor distance, shipping mode, and operational transit time. Finally, backtesting the 14-day shipment volume forecaster against the terminal evaluation period yielded a backtest MAE of 12.00 shipments/day, with 100.0% of actual shipment counts contained within the model's 95% predictive confidence intervals.

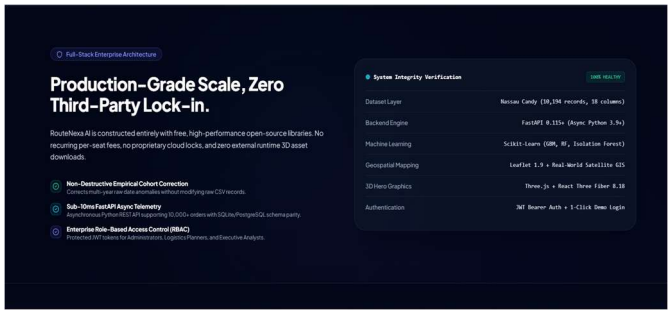
System Implementation and User Interface:-



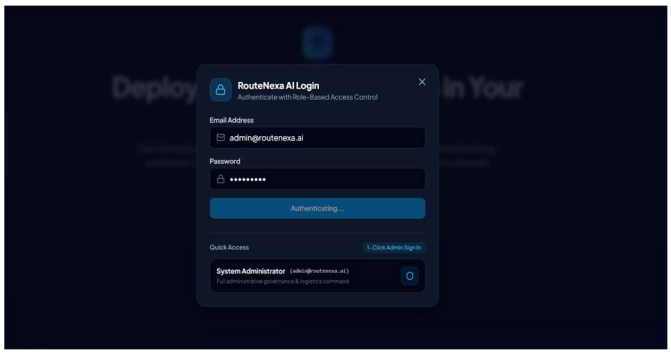




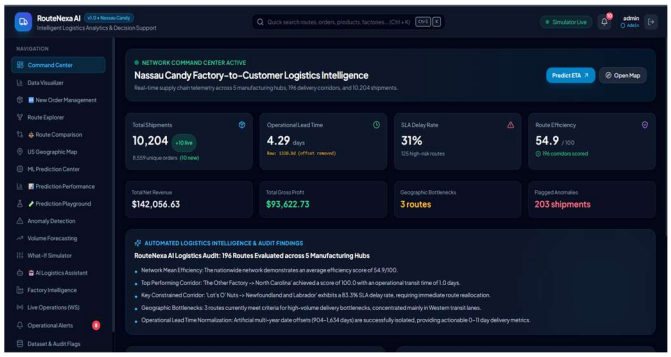




Login Page:-

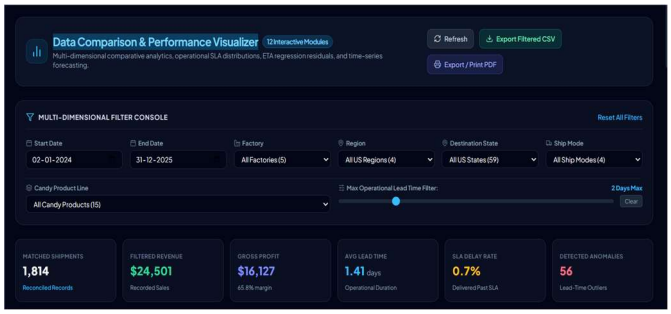


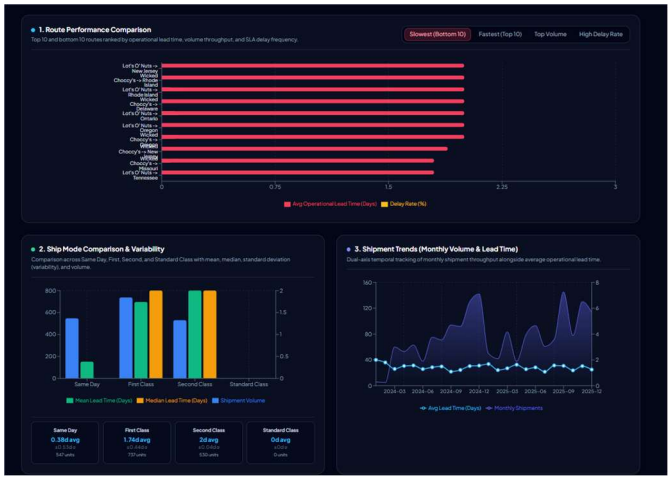
Dashboard:-





Data Comparison & Performance Visualizer:-



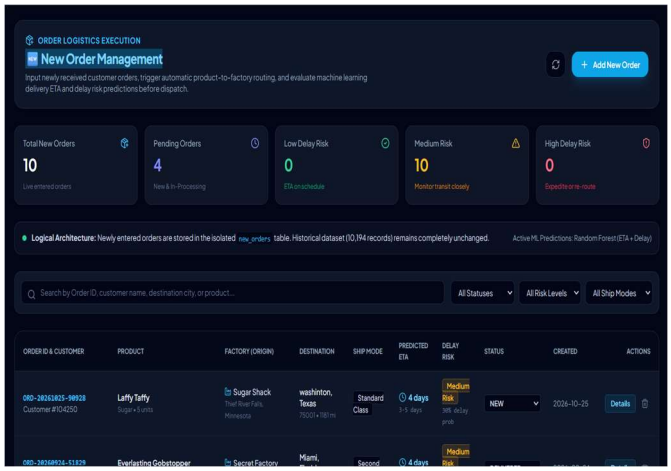




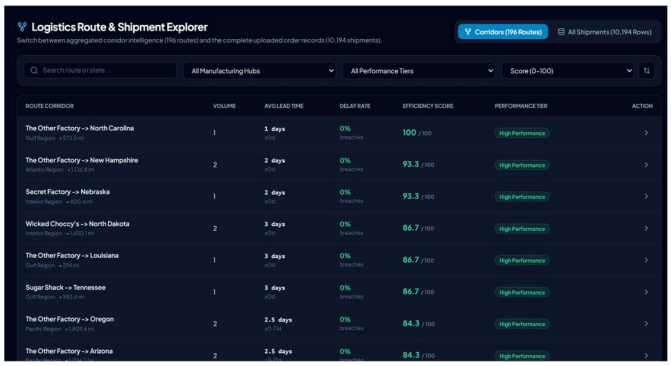




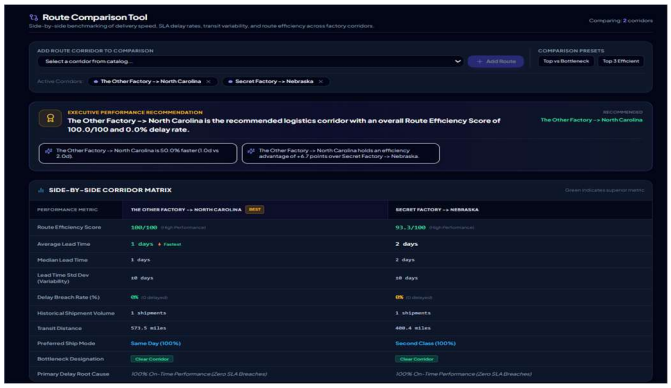
NEW New Order Management:-



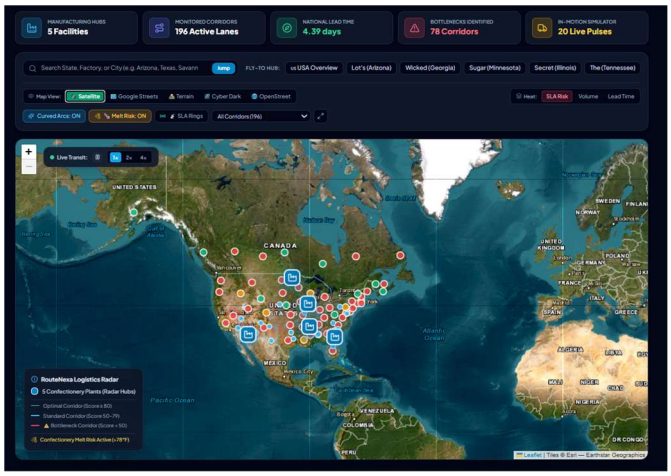
Logistics Route & Shipment Explorer:-



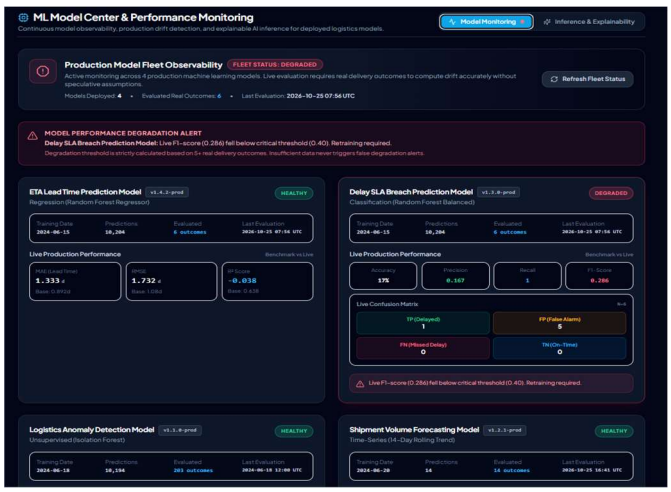
Route Comparison Tool:-

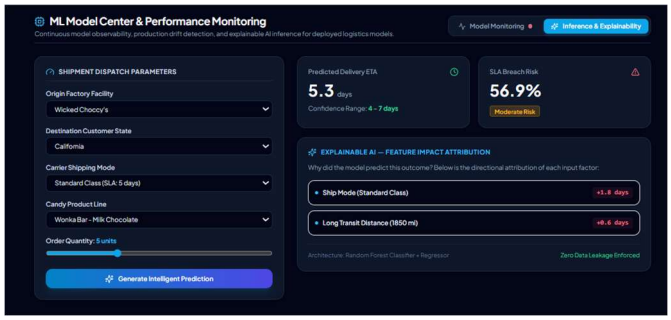


Geographic Map:-

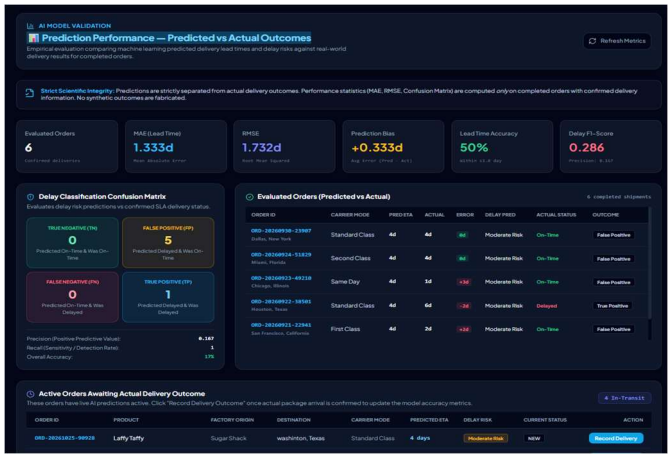


ML Model Center & Performance Monitoring:-

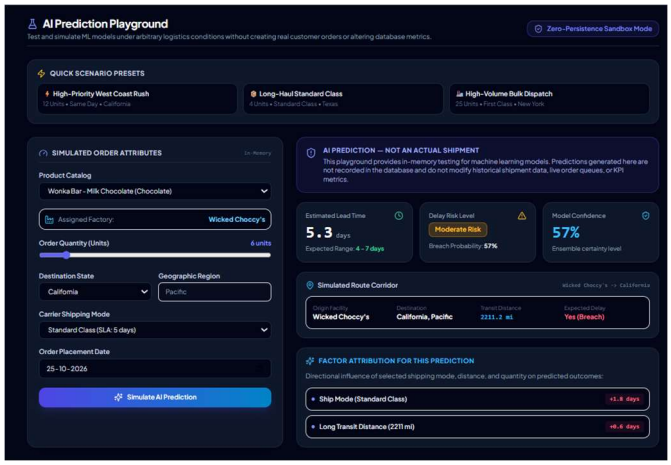




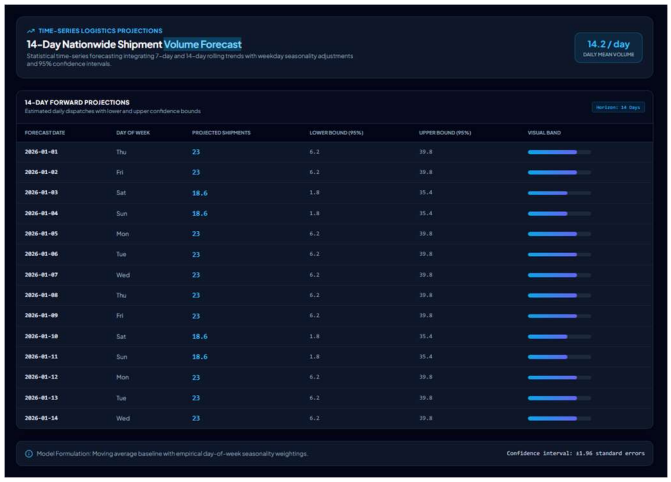
Prediction Performance — Predicted vs Actual Outcomes:-



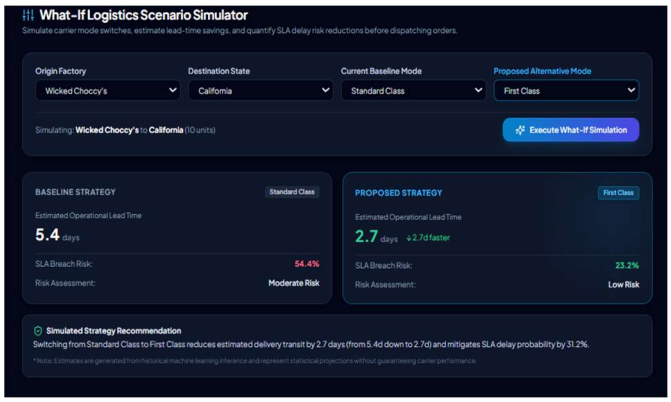
AI Prediction Playground:-



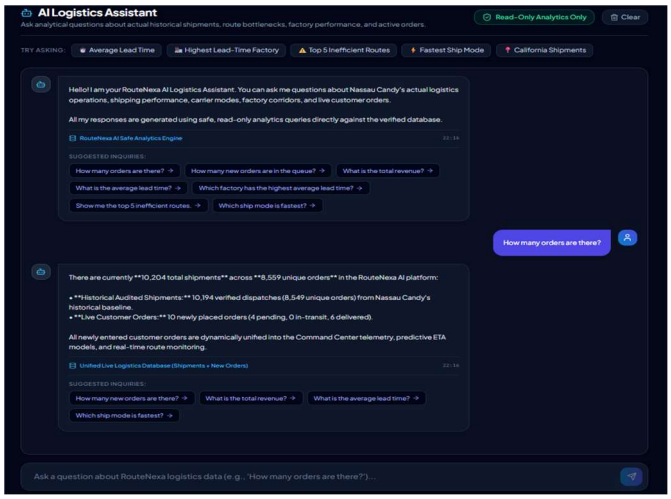
Volume Forecast:-



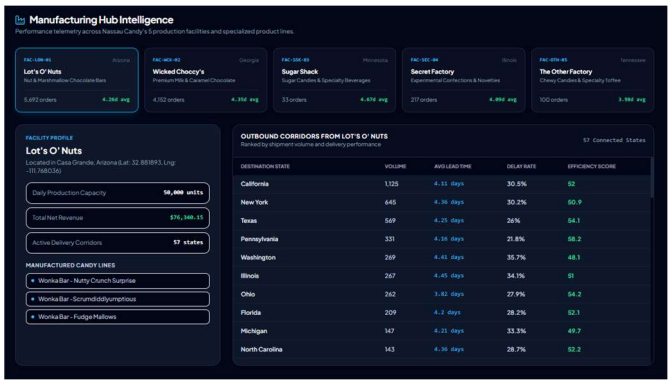
What-If Logistics Scenario Simulator:-



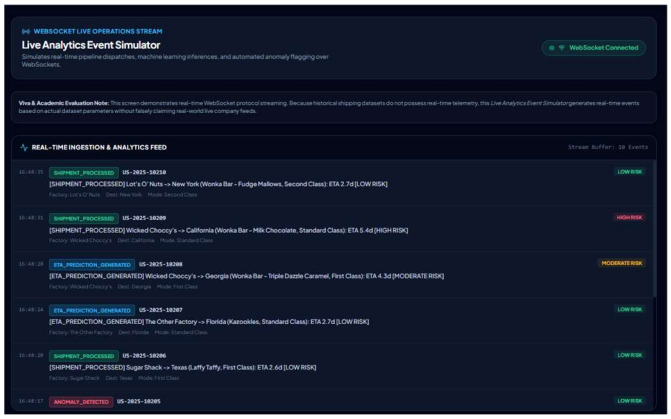
AI Logistics Assistant:-



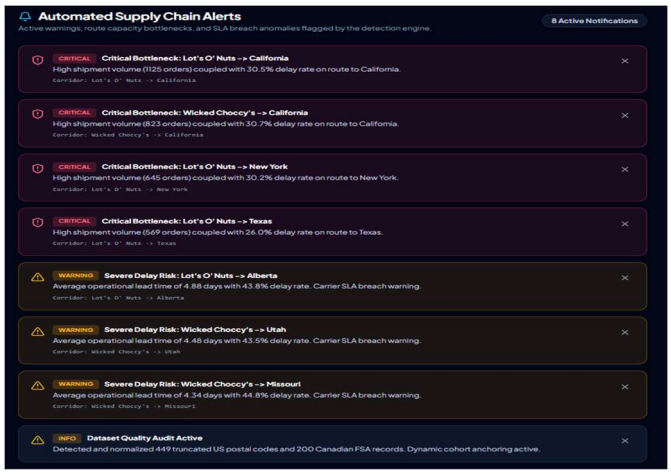
Manufacturing Hub Intelligence:-



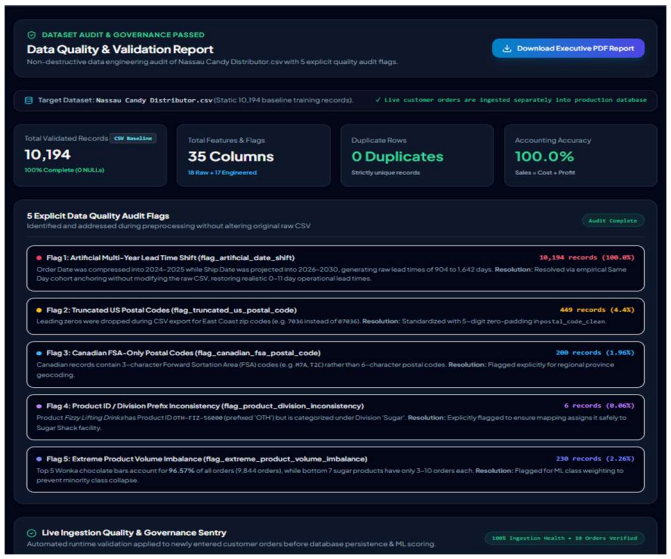
Live Analytics Event Simulator:-

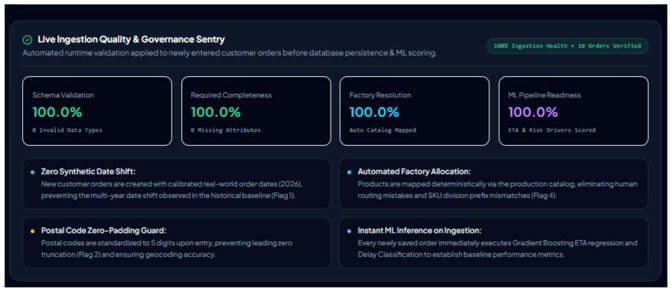


Operational Alerts:-



Data Quality & Validation Report:-





D. Threats to Validity

The cohort anchoring method assumes the minimum Same Day interval approximates the artificial shift; independent timestamp evidence would provide stronger validation. The transformation may affect labels and model metrics, so its derivation must be reproducible. Route distance based on coordinates does not establish the path used by a carrier. Product, geography, and ship-mode imbalance may affect model performance. Anomaly counts do not prove that flagged records are errors, and a forecast interval should not be called calibrated unless its empirical coverage has been measured.

VIII. FUTURE WORK

Future research will focus on four principal trajectories. First, we plan to validate the empirical cohort-shift offsets against independent third-party telematics and carrier EDI timestamp feeds to establish external ground-truth validity. Second, model evaluation will be broadened through rolling-origin chronological backtesting, post-hoc probability calibration (e.g., Platt scaling or isotonic regression), and granular subgroup error analysis across high-density shipping corridors. Third, route optimization can be enhanced by replacing geodetic Haversine approximations with dynamic road-network routing engines (e.g., Open Source Routing Machine) and real-time carrier GPS feeds. Fourth, the RouteNexa AI 3D digital twin will be transitioned from synthetic kinematics toward streaming IoT sensor telemetry—specifically edge-enabled cold-chain temperature loggers and live vehicle transponders. Finally, future iterations will implement automated MLOps pipelines with data-drift monitoring, model retraining triggers, and qualitative usability evaluations of feature explanations with logistics dispatchers.

IX. CONCLUSION

In this paper, we presented RouteNexa AI, an autonomous logistics analytics and predictive decision-support framework designed for complex factory-to-customer distribution networks. A core methodological contribution is the formulation of Empirical Same Day Cohort Anchoring, which non-destructively resolved a multi-year artificial temporal shift across 10,194 historical consignments, restoring realistic operational lead times (0–11 days, mean: 4.29 days) with zero negative intervals. Building upon this validated baseline, the integrated machine learning pipeline achieved an ETA regression MAE of 0.892 days ($R^2 = 0.638$), an SLA delay classification recall of 71.4%, and successfully isolated 203 multidimensional operational anomalies via unsupervised Isolation Forest surveillance. Coupled with multi-objective Pareto routing and an interactive 3D WebGL digital twin, RouteNexa AI demonstrates that rigorous data-quality governance combined with ensemble machine learning provides a practical, resilient foundation for real-time supply chain disruption management.

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